SUMMARY TABLE: PHYSICS GRADE 10

Sub-Strand 3.1: Radioactivity and Stability of Isotopes — Teacher Reference (Pre-filled)

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Sub-Strand	Sub-Strand 3.1: Radioactivity and Stability of Isotopes
Driving Question	DRIVING QUESTION: "Why do some atoms in Kenyan rocks spontaneously release invisible energy — and how can we harness or protect ourselves from that energy?"

INSTRUCTIONS
FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 **The Invisible Killer in the Rock: Nuclear Structure, Isotopes & Launching the Phenomenon**	Students observed (or were shown) a demonstration in which a sample of pitchblende (or a photo of uranium-bearing rock found in Kenya's geological surveys) was placed near a covered photographic film or a Geiger counter, producing a reading/exposure with no visible light source. This anchoring phenomenon — invisible energy escaping from a rock — launched the Driving Question Board. Pedagogical note: If a live source is unavailable, a time-lapse image of film fogging beside pitchblende works well. Allow 3–5 minutes of unguided observation before asking any questions; record all student 'noticings' and 'wonderings' verbatim on the DQB.	Students learned that every atom has a nucleus composed of protons (atomic number Z) and neutrons, collectively called nucleons (mass number A). The neutron number is derived as $N = A - Z$. Isotopes are atoms of the same element sharing identical Z but differing in N and therefore in A (e.g., Carbon-12 vs Carbon-14; Uranium-235 vs Uranium-238). Nuclide notation ᴬ_Z X was introduced as the standard way to identify any nuclide. Students learned that some isotopes are stable while others are unstable (radioactive), and that nuclear instability — not chemical bonding — is the source of the invisible energy in the anchoring phenomenon. Pedagogical note: Emphasise that 'isotope' is a relationship between two nuclides of the same element; 'nuclide' refers to any specific nuclear species. Misconception alert — many students confuse atomic mass with mass number; clarify that atomic mass is a weighted average, while mass number is always a whole number.	Students explained that the pitchblende rock releases invisible energy because it contains radioactive isotopes (primarily U-238 and U-235) whose nuclei are unstable. The instability arises from an unfavourable proton-to-neutron ratio in the nucleus. Because the nucleus is trying to reach a more stable configuration, it spontaneously releases energy and particles — a process called radioactive decay. This explanation directly answers the first clause of the driving question ('Why do some atoms spontaneously release invisible energy?') at an introductory level, and sets up subsequent lessons to explore what forms that energy takes, how it is detected, and how it can be harnessed or blocked. Pedagogical note: Return to the DQB and annotate which student questions from the 'wondering' column are now partially answered, and which remain open — this models scientific sensemaking and sustains curiosity.
Lesson 2	Students observed a teacher-led	Students learned that radioactive nuclei	Students explained how each radiation type

Types of Radiation: Alpha, Beta, and Gamma — Penetrating Power and Nuclear Identity	demonstration (or simulation) comparing the penetration of alpha, beta, and gamma radiation through paper, aluminium foil, and lead sheets. Geiger counter readings (live or pre-recorded) were recorded in a results table for each material-radiation combination. A Kenya-context analogy was drawn: alpha particles are like a matatu that stops at the first stage; beta particles travel further like a Nairobi express bus; gamma rays pass through everything like a radio signal from a Kisumu transmitter. Pedagogical note: If physical sources are unavailable, use a PhET-style or ARES simulation. Ensure students record quantitative Geiger readings (counts per minute) so the comparison is evidence-based, not just descriptive.	emit three distinct types of radiation. Alpha radiation (α) consists of a helium-4 nucleus (charge +2, mass ≈ 4 amu); it has the highest ionising power but the lowest penetrating power (stopped by a sheet of paper or a few centimetres of air). Beta radiation (β) is a fast-moving electron (charge −1, mass ≈ 1/1836 amu) emitted when a neutron converts to a proton; it has intermediate ionising power and penetrating power (stopped by a few millimetres of aluminium). Gamma radiation (γ) is high-energy electromagnetic radiation (charge 0, no mass); it has the lowest ionising power but the highest penetrating power (requires centimetres of lead or metres of concrete to attenuate significantly). Pedagogical note: Stress that penetrating power and ionising power are inversely related — this is a frequently tested relationship. Use a two-column comparison table on the board and have students copy it into their notebooks.	relates to the driving question's anchoring phenomenon: the 'invisible energy' from Kenyan rocks is a mixture of alpha, beta, and gamma emissions from unstable isotopes like U-238 and its decay products. The different penetrating powers explain why some radiation is absorbed within the rock itself (alpha) while gamma rays can travel through the rock and into the surrounding environment — making gamma the primary external hazard to people living near uranium-bearing geological formations in the Rift Valley. Students also explained that the type of radiation emitted is determined by the specific instability of the nucleus, connecting back to Lesson 1's discussion of unfavourable proton-to-neutron ratios. Pedagogical note: Cold-call at least three students to explain why alpha radiation, despite being the most ionising, poses the greatest biological danger when a source is ingested rather than external — this checks for transfer of understanding.
Lesson 3 **Detection of Radiation: Seeing the Invisible**	Students observed either a live Geiger-Müller (GM) tube connected to a counter/ratemeter, or a detailed labelled diagram and video demonstration. Background radiation readings (counts per minute) were recorded first, then readings with a source present. Students also examined (or were shown images of) cloud chamber tracks, with particular attention to the different track thicknesses corresponding to alpha (thick, straight) and beta (thin, curved) particles. Pedagogical note: Always measure and record background radiation before introducing any source — this is both good experimental practice and an important teaching point about natural background radiation from Kenyan soil, rocks, and cosmic rays.	Students learned that: (1) A Geiger-Müller tube detects radiation because ionising radiation enters through a thin mica window, ionises argon gas atoms inside the tube, producing a cascade of electrons that generates a measurable electrical pulse counted as 'clicks' or counts per minute (cpm). (2) Background radiation is always present (from cosmic rays, radon gas in soil, building materials, food such as bananas which contain K-40) and must be subtracted from all experimental readings to obtain the corrected count rate from a source. (3) A cloud chamber makes radiation tracks visible by supersaturating alcohol vapour — ionising radiation creates ion trails along which droplets condense. (4) Photographic film darkens upon exposure to radiation — this is exactly what happened to the pitchblende in Lesson 1's anchoring phenomenon. Pedagogical note: The link between the film-fogging in Lesson 1 and photographic film as a detector is a critical conceptual anchor — explicitly revisit	Students explained that the reason the pitchblende rock in Lesson 1 fogged the photographic film is precisely the same ionising mechanism that makes a GM tube click: radiation from unstable nuclei ionises matter it passes through. This explains how Marie Curie (and Kenyan geologists surveying the Rift Valley today) can 'see' invisible radiation — by observing its ionising effects on matter. Students further explained that background radiation means people in Kenya are always exposed to low levels of nuclear radiation from natural sources, and that detectors allow scientists to distinguish between safe background levels and dangerous elevated levels near radioactive rock formations. Pedagogical note: Reinforce the SEP (Science and Engineering Practice) of 'Using Mathematics and Computational Thinking' when students subtract background count rates — insist on correct units (cpm) and proper significant figures.

		the Lesson 1 phenomenon here.	
Lesson 4 **Radioactive Decay and Half-Life: The Rock's Countdown Clock**	Students observed a hands-on simulation of radioactive decay using 100 coins (or 100 dried maize kernels, a Kenya-relevant manipulative): each 'shake-and-remove-heads' trial represents one half-life. Results were recorded in a table (number of 'undecayed' kernels vs. trial number) and plotted as a decay curve. Alternatively, students analysed a pre-drawn decay curve for Carbon-14 (used in dating organic archaeological finds at sites like Olorgesailie in Kenya) and read off half-life values graphically. Pedagogical note: The maize-kernel analogy is culturally resonant and memorable. Emphasise that each individual kernel's 'decay' is random and unpredictable, but the overall pattern is statistically regular — this is the quantum-probabilistic nature of radioactive decay. Allow students to articulate this distinction before providing the formal definition.	Students learned that: (1) Radioactive decay is a random and spontaneous process — it is impossible to predict when any single nucleus will decay, but the decay of a large population of nuclei follows a precise statistical law. (2) The half-life ($t_{1/2}$) is the time taken for half the nuclei in a sample to decay (equivalently, for the activity of a sample to halve). Half-life is a characteristic constant for each radioisotope and is unaffected by temperature, pressure, or chemical state. (3) After n half-lives, the fraction of original nuclei remaining is $(1/2)^n$. (4) Half-lives span an enormous range: Polonium-214 has $t_{1/2} \approx 164$ μs; Carbon-14 has $t_{1/2} \approx 5{,}730$ years; Uranium-238 has $t_{1/2} \approx 4.5$ billion years. (5) Activity (measured in Becquerels, Bq) decreases in the same pattern as the number of undecayed nuclei. Pedagogical note: Spend time on the graph — students must be able to read $t_{1/2}$ from a decay curve, calculate remaining quantity after multiple half-lives, and work backwards to find elapsed time. These are high-frequency exam skills.	Students explained that the long half-life of Uranium-238 (4.5 billion years, comparable to the age of Earth) is why Kenyan rocks still contain enough radioactive material to pose a measurable signal today — if the half-life were short (like Polonium-214), all the radioactivity would have vanished billions of years ago. Conversely, the shorter half-lives of decay products in the uranium decay chain mean some daughter isotopes are far more active (higher activity) than U-238 itself, contributing disproportionately to the radiation hazard in uranium-bearing rocks. Students also connected half-life to the driving question's 'harness' clause: knowing the half-life of a medical radioisotope (e.g., Technetium-99m, $t_{1/2} \approx 6$ hours, used in Kenyatta National Hospital imaging) allows doctors to calculate safe dosing windows. Pedagogical note: Update the DQB — students should now be able to answer 'How long will the rock keep releasing energy?' with quantitative reasoning.
Lesson 5 **Nuclear Equations: Writing and Balancing Radioactive Decay**	Students observed worked examples of alpha and beta decay written as nuclear equations on the board (e.g., U-238 → Th-234 + α; C-14 → N-14 + β^-). They then participated in a card-sorting activity (or whiteboard practice) in which nuclide notation cards were arranged to balance equations, making conservation of A and Z a physical, manipulable process before it became an abstract rule. Pedagogical note: Insist students check BOTH conservation conditions (ΣA and ΣZ) explicitly for every equation they write — a two-row checking grid drawn under each equation (row 1: A values, row 2: Z values) is an effective scaffold that prevents the most common errors.	Students learned that: (1) Nuclear equations must conserve nucleon number (A) and proton number (Z) on both sides — these are the fundamental conservation laws governing all radioactive decay. (2) In alpha decay, A decreases by 4 and Z decreases by 2, producing a daughter nucleus two elements lower in the periodic table. (3) In beta-minus (β^-) decay, A stays constant while Z increases by 1 (a neutron converts to a proton plus an electron plus an antineutrino); the daughter nucleus is one element higher. (4) In gamma emission, neither A nor Z changes — the nucleus merely releases excess energy as a gamma photon. (5) Nuclear equations use the full nuclide notation $^{A}_{Z}X$ for every particle, including the emitted radiation (e.g., $^{4}_{2}He$ for alpha, $^{0}_{-1}e$ for beta).	Students explained that balancing nuclear equations is the mathematical language that describes the transformation taking place inside the Kenyan rock's unstable nuclei. Each decay step in the U-238 decay chain (which passes through 14 intermediate nuclides before reaching stable Pb-206) can be written as a balanced nuclear equation. This chain explains why uranium-bearing rocks produce a whole family of radioactive daughters — including Radon-222, a gas that can seep into poorly ventilated homes built on uranium-rich ground in parts of Kenya, posing a lung-cancer risk. Writing and balancing the equation for Rn-222 alpha decay connects abstract nuclear arithmetic directly back to the driving question's 'protect ourselves' clause. Pedagogical note: Having students

		Pedagogical note: Beta-plus (positron) decay is not required at this level per KICD, but high-ability students may ask about it — a brief acknowledgement with a note that it is beyond the current syllabus is appropriate.	trace at least two steps of the U-238 decay chain using balanced equations is an excellent formative assessment task that integrates Lessons 1, 2, and 5.
Lesson 6 **Nuclear Fission and Fusion: Unlocking the Energy Inside the Rock**	Students observed a diagram or animation of nuclear fission (U-235 splitting upon neutron absorption, releasing fission fragments, 2–3 neutrons, and energy) and nuclear fusion (two hydrogen isotopes combining to form helium, releasing even greater energy). A Kenya-context entry point was used: 'Kenya imports petroleum — could nuclear energy ever replace it?' Students examined data comparing energy yields per kilogram of fuel for coal, petroleum, and uranium fission, and noted that 1 kg of U-235 yields energy equivalent to approximately 3,000 tonnes of coal. Pedagogical note: Use a quantitative data table to make the energy comparison concrete. The chain reaction diagram — showing one neutron triggering fission that releases three neutrons, each triggering further fissions — is essential for understanding both reactors and the concept of critical mass.	Students learned that: (1) Nuclear fission is the splitting of a heavy nucleus (e.g., U-235) by an incoming neutron, producing two smaller fission fragment nuclei, 2–3 free neutrons, gamma radiation, and a large release of energy (≈200 MeV per fission event) explained by Einstein's mass-energy equivalence $E = mc^2$. (2) The released neutrons can trigger further fission events, establishing a chain reaction; in a nuclear reactor, control rods (made of boron or cadmium) absorb excess neutrons to maintain a controlled, sustained chain reaction. (3) Nuclear fusion is the joining of two light nuclei (e.g., deuterium + tritium → helium-4 + neutron) at extremely high temperatures (≈10^7 K), releasing even more energy per unit mass than fission; fusion powers the Sun and all stars. (4) Mass defect (Δm) is the difference between the mass of reactants and products in a nuclear reaction; this 'missing' mass is converted to energy via $E = mc^2$. Pedagogical note: Students should be able to state $E = mc^2$ and explain qualitatively (not calculate, unless required by the scheme of work) that a tiny mass converts to an enormous energy due to $c^2 \approx 9 \times 10^{16}$ m²/s².	Students explained that the energy released by Kenyan uranium-bearing rocks through natural radioactive decay is the same fundamental phenomenon — mass converting to energy — as occurs in nuclear fission reactors and stellar fusion, just at a far smaller scale and rate. This directly answers the driving question's 'harness' clause: Kenya could in principle use uranium from its own geological formations as fuel in a nuclear power reactor to generate electricity without burning fossil fuels, reducing dependence on petroleum imports and hydroelectric sources that are vulnerable to drought (as seen during Kenyan power-rationing periods). Students also noted that fusion, while not yet commercially viable globally, represents an even cleaner future energy source. Pedagogical note: A structured class debate — 'Should Kenya build a nuclear power plant?' — is highly effective here as a PCIs-aligned activity (social cohesion, citizenship) and ensures students synthesise energy yield, safety, and environmental considerations.
Lesson 7 **Applications, Safety, and Unit Review: Closing the Case of the Invisible Killer**	Students revisited the original anchoring phenomenon from Lesson 1 (the pitchblende rock fogging photographic film) and, using all accumulated knowledge, constructed a full written and diagrammatic explanation. They also examined a case-study card set presenting real-world applications of radioactivity in a Kenyan context: (a) Cobalt-60 radiotherapy at Kenyatta National Hospital; (b) Carbon-14 dating of archaeological artefacts at Olorgesailie; (c) food irradiation to extend	Students learned that the same nuclear instability that caused the pitchblende rock to expose photographic film in Lesson 1 is the foundational phenomenon behind a wide spectrum of beneficial applications and serious hazards: (1) Medical: gamma radiation from Co-60 destroys cancer tumours (radiotherapy); radioactive tracers (e.g., Tc-99m) enable imaging of organs at KNH. (2) Archaeological: C-14 dating exploits the known half-life (5,730 yr) of Carbon-14 to determine the age of organic	Students synthesised the full arc of the sub-strand: unstable isotopes in Kenyan rocks (rooted in unfavourable proton-to-neutron ratios, Lesson 1) spontaneously emit alpha, beta, and gamma radiation (Lesson 2) that can be detected by GM tubes, film, and cloud chambers (Lesson 3). This decay follows a precise half-life law (Lesson 4), describable by balanced nuclear equations (Lesson 5), and involves mass-to-energy conversion also seen in fission and fusion (Lesson 6). The same phenomenon that

	shelf life of Kenyan horticultural exports (e.g., Rift Valley green beans); (d) industrial thickness gauging in Kenyan cement and paper factories; (e) radon gas risk in homes built on Kenyan granite. Students matched each application to the relevant radiation type, half-life consideration, and safety measure. Pedagogical note: The case-study matching activity serves as a rich formative assessment. Circulate and listen for misconceptions — the most common is that irradiated food becomes radioactive; address this explicitly.	remains. (3) Agricultural/industrial: gamma irradiation sterilises food and medical equipment; beta sources gauge material thickness in manufacturing. (4) Energy: nuclear fission of U-235 generates electricity in nuclear power stations with zero carbon emissions during operation. (5) Hazards and safety: excessive radiation causes ionisation damage to DNA, leading to cancer or cell death; safety measures include distance (inverse-square law), shielding (lead/concrete for gamma, aluminium for beta, paper/skin for alpha), and minimising exposure time. Radiation workers wear film badges or TLD dosimeters; international dose limits are set in Sieverts (Sv). Pedagogical note: Summarise the entire sub-strand by returning explicitly to the DQB from Lesson 1 — every question posted should now be answerable. Demonstrate this publicly by reading each question aloud and cold-calling students to provide answers, annotating the board as a class.	makes uranium rocks hazardous also makes nuclear medicine, carbon dating, food preservation, and potentially nuclear electricity generation possible in Kenya (Lesson 7). The driving question — 'Why do some atoms in Kenyan rocks spontaneously release invisible energy, and how can we harness or protect ourselves from that energy?' — is now fully answered: atoms release energy to reach nuclear stability, and humans manage that energy through informed application of physics principles including shielding, distance, half-life knowledge, and controlled chain reactions. Pedagogical note: A strong closing formative task is to ask each student to write a one-paragraph 'letter to a Kenyan farmer living near a uranium-bearing rock formation in the Rift Valley' explaining the risk and what protective steps to take — this integrates scientific literacy, communication competency, and Kenya-context relevance in a single authentic task.